

Smart Logistics Command Center - Project Report

Technology Stack: Spring Boot, MongoDB, React, Vite, JWT Authentication

1. Project Overview

Smart Logistics Command Center is a full-stack logistics management system for managing warehouses, products, vehicles, drivers, shipments, alerts, analytics, audit logs, and MongoDB advanced features.

Development Environment

Tool / IDE	Version
Java JDK	25
Spring Boot	3.x
Maven	3.9+
MongoDB Community Server	8.x
MongoDB Compass	Latest
Mongo Shell (mongosh)	2.x+
Node.js	22.x LTS
npm	10.x+
React	19.x
Vite	7.x
Axios	1.x
Recharts	3.x
VS Code	Latest
Postman	Latest
Google Chrome	Latest

Backend Technologies

Spring Boot, Spring Web, Spring Data MongoDB, Spring Security, JWT Authentication

Frontend Technologies

React, Vite, Axios, React Router DOM, Recharts

Database Technologies

MongoDB Collections, Aggregation Pipelines, Indexing, Explain Plans, Replica Sets

2. Features Implemented

- JWT Authentication
- Warehouse Management
- Product Management
- Vehicle Management
- Driver Management
- Shipment Management

- Risk Prediction
- Alerts Dashboard
- Analytics Dashboard
- Audit Logging
- MongoDB Aggregations
- MongoDB Indexing
- MongoDB Replica Set

3. MongoDB Collections

- users
- warehouses
- products
- vehicles
- drivers
- shipments
- alerts
- auditLogs

4. API Endpoints

Method	Endpoint
POST	/api/auth/login
GET	/api/warehouses
POST	/api/warehouses
DELETE	/api/warehouses/{id}
GET	/api/products
POST	/api/products
DELETE	/api/products/{id}
GET	/api/vehicles
POST	/api/vehicles
DELETE	/api/vehicles/{id}
GET	/api/drivers
POST	/api/drivers
DELETE	/api/drivers/{id}
GET	/api/shipments
POST	/api/shipments
DELETE	/api/shipments/{id}
GET	/api/alerts
GET	/api/alerts/risk/{risk}
GET	/api/analytics/shipment-status
GET	/api/analytics/risk-distribution
GET	/api/audit

5. MongoDB Features

Aggregation Pipelines for shipment status and risk distribution analytics.

Indexes for optimized shipment search using shipmentId.

Explain Plan verification using IXSCAN.

Replica Set configuration using three MongoDB nodes.

Automatic demo data loading using DataLoader.

6. Default Login

Username: admin

Password: admin

7. How to Run

1. Start MongoDB
2. Run Spring Boot backend
3. Run React frontend
4. Login using admin/admin

8. Conclusion

The project demonstrates a complete logistics management platform using MongoDB with CRUD operations, analytics, indexing, audit logging, JWT authentication, and replica set support.